Exact and Structural Methods for the Job Sequencing and Tool Switching Problem

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Abstract

The Job Sequencing and Tool Switching Problem (SSP) seeks an ordering of jobs on a single flexible machine, together with a tool-loading policy for a capacitated magazine, that minimises the number of tool switches. For a fixed order the loading is solved exactly by a classical greedy rule, so the difficulty lies in the ordering, and the problem is NP-hard. We study the SSP through its configuration view—jobs as clusters of magazine-filling tool sets—along two lines. The first asks how far the standard group-then-sequence heuristic can be from optimal: we prove that the SSP optimum equals the minimum configuration-walk cost over *all* feasible groupings, so that the heuristic's gap is exactly the price of using a minimum-cardinality grouping; we then give matching lower bounds, a family on which the gap grows without bound, a worst-case ratio guarantee of $\min(b,\ K^*-1,\ |U|-b)$ together with zero-gap classes valid for every capacity, exact and quantitative gap laws at $K^*=3$ that are tight at every known extremal instance, a refutation of the natural K^*-2 gap conjecture, and a proof of the 4/3 ratio bound at $b=3$ whenever $K^* \leq 3$. The second develops exact methods—a branch-and-Benders-cut algorithm that exploits the polynomial loading subproblem, and position-indexed formulations (PCF, PCF′, PTF) with a fixed row set and a branch-and-price. A single quantity, the coverage bound $|U|-b$, links the two: it is the value every configuration relaxation attains and, as the gap analysis shows, the exact optimum on a large part of the instance space—so it both bounds the heuristic and limits configuration-based exact methods. Results proved here are stated as theorems; those resting on enumeration are stated as conjectures; the large-scale benchmark is in progress.

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1 Introduction

Motivation. A flexible manufacturing cell built around a single computer-numerically-controlled machine can produce a wide variety of parts because it draws on a large library of cutting tools. The machine, however, holds only a limited number of tools at once, in a magazine of capacity b; producing the full part mix therefore requires tools to be exchanged between jobs. Each exchange interrupts production while the magazine is reconfigured, so tool switches are the dominant controllable cost in such cells [11]. When the set of tools needed by each job is fixed, two decisions remain: in what *order* to process the jobs, and which tools to *keep or evict* at each step. The Job Sequencing and Tool Switching Problem (SSP) is the joint optimisation of these two decisions so as to minimise the total number of switches. The same structure—a capacitated store serving set-valued requests that may be reordered—recurs beyond machining, for example in printed-circuit-board assembly and in warehouse order picking, so the methods developed here are not specific to tools.

Why the problem is subtle. The two decisions are coupled in opposite directions. For a *fixed* job order, the optimal loading policy is known in closed form—the Keep Tool Needed Soonest (KTNS) rule of Tang and Denardo [32], computable in $O(n|T|)$ time—so the loading subproblem is easy. The difficulty lies entirely in the ordering: a good order places jobs with similar tool requirements next to one another so that few tools change hands, but "similarity" here is a set-overlap relation with no underlying metric, and the capacity constraint couples positions that are far apart in the sequence. The joint problem is NP-hard [11, 20], and even the strongest compact integer formulations [6, 12, 20] become difficult beyond moderate sizes. This structure—an easy inner problem nested inside a hard ordering problem—is what decomposition methods are designed to exploit, and both lines of this work, the structural analysis and the exact methods, rest on it.

The grouping view, and the questions it forces. The SSP is closely tied to a simpler relative, the Job Grouping Problem (JGP), which asks only how few magazine configurations suffice to serve every job. Grouping suggests both the standard heuristic—group the jobs, then order the groups—and a route to exact solution, because a *configuration* (a magazine-filling set of tools) is a natural unit to reason about: the SSP becomes a generalised travelling-salesman problem that visits, in some order, one configuration able to serve each job, paying for the tools that change between consecutive configurations [8]. Two questions follow, and the literature answers neither. First, how good is the heuristic? Group-then-sequence is the method of choice in applications—in colour printing, where the SSP appears verbatim, it comes close to optimal on industrial instances [5], and the strongest metaheuristics seed from JGP solutions [22]—but its record is entirely empirical: how far it can be from the optimum, and when it is exact, has not been analysed. Second, how well can the SSP be solved exactly through configurations—branch-and-price is effective for the JGP because its set-covering relaxation is tight [27], so does the configuration view yield an equally strong exact method for the SSP, or is it blocked because the natural configuration relaxation collapses to the elementary bound $|U| - b$?

A single quantity links the two. These questions share an answer in that coverage bound $|U| - b$, the number of used tools in excess of the magazine capacity. It is at once the value every configuration relaxation attains and, as the gap analysis shows, the exact optimum on a large part of the instance space. Where it is tight, exact configuration methods close at the root and the heuristic tends to be exact; where it is loose, the heuristic's gap can open and the same methods branch heavily. That coincidence, developed in Sections 3 and 5, is the connecting thread of this report, and the configuration view that produces it is established once, in Section 2.

Contributions. The principal results are the following.

- *A reframing of the heuristic gap* (Section 3): we prove that the SSP optimum equals the minimum GTSP cost over *all* feasible groupings, not only minimum-cardinality ones. Consequently the two-phase heuristic's optimality gap is exactly the cost of restricting attention to minimum-cardinality groupings.
- *Gap quantification*: we prove the matching lower bounds $Z^* \geq \max(K^* - 1, |U| - b)$, the worst-case guarantee $H/Z^* \leq \min(b,\ K^* - 1,\ |U| - b)$, and zero-gap classes valid for every b ($K^* \leq 2$, or $|U| \leq b + 1$), an exact expression for the heuristic cost at $K^* = 3$, and quantitative gap bounds for every K^*; we exhibit a family on which the gap grows linearly without bound; we *refute* the natural conjecture gap $\leq K^* - 2$ by an explicit $K^* = 3$, $b = 5$ counterexample lying exactly on the boundary of the proved regimes and attaining our bounds with equality; and we prove the 4/3 ratio bound at $b = 3$ for every $K^* \leq 3$, stating the remaining $K^* \geq 4$ case as the one surviving conjecture.
- *Structure of the gap* (Section 3.4): the feasible groups form an independence system whose circuit clutter computes K^* as a hypergraph chromatic number, yet provably does not determine the gap—two instances with identical circuit structure and equal (K^*, Z^*) have different gaps—so any zero-gap criterion must read tool-level overlaps, as the exact $K^* = 3$ expression does.
- *A collapse theorem*: under a per-changeover magazine setup cost, the heuristic becomes exact once the ratio of setup cost to switch cost exceeds an explicit threshold.
- *Exact methods* (Section 4): a branch-and-Benders-cut solver whose subproblem is the polynomial KTNS oracle; and position-indexed formulations (PCF, its symmetry-reduced variant PCF′, and the transition model PTF) with a fixed $O(n|T|)$ row set, their pricing subproblems, and an exact branch-and-price.
- *An objective competitiveness assessment* (Section 5): for the SSP the configuration relaxation collapses to the trivial coverage bound on most instances, so a configuration-based

branch-and-price is bound-limited rather than implementation-limited.

Outline. Section 2 fixes notation, defines the SSP and KTNS, introduces the two-phase heuristic and the configuration–GTSP lens, proves the two structural lower bounds, fixes the switch-cost convention, and reviews the literature. Section 3 develops the structural theory of the heuristic gap; Section 4 presents the exact methods; Section 5 reports the experimental design and the competitiveness assessment; Section 6 concludes; and Section 7 sets out the work planned for the remainder of the project, phased by dependency. Longer proofs are deferred to Appendix A.

2 Preliminaries

2.1 The Job Sequencing and Tool Switching Problem

We are given n jobs $J = \{1, \ldots, n\}$, m tools $T = \{1, \ldots, m\}$, a magazine capacity $b \in \mathbb{N}$ with $b < m$, and for each job j a set of required tools $T_j \subseteq T$ with $1 \leq |T_j| \leq b$. We write $U = \bigcup_{j \in J} T_j$ for the set of tools used by at least one job.

Definition 1 (Schedule and switch cost)**.** A *schedule* is a pair $(\pi, \mathbf{M})$ consisting of a permutation π of J and a sequence of *magazine states* $\mathbf{M} = (M_1, \ldots, M_n)$, $M_i \subseteq T$, satisfying the capacity and feasibility constraints

$$|M_i| \leq b \quad \text{and} \quad T_{\pi(i)} \subseteq M_i \qquad (i = 1, \ldots, n).$$

With $M_0 = \varnothing$, its *switch cost* is the total number of tool insertions, $\sum_{i=1}^{n} |M_i \setminus M_{i-1}|$. The SSP asks for a schedule of minimum switch cost; this optimum is denoted Z^*.

Counting insertions is the standard cost model: under a fixed capacity each insertion into a full magazine forces exactly one removal, so insertions and removals differ only by boundary terms, which Section 2.6 makes precise.

Example 1 (A running instance: the 6-ring)*.* We illustrate every definition on one instance, returned to throughout (Figure 1). The 6*-ring* has $n = m = 6$, capacity $b = 3$, and $T_j = \{j, j+1\}$ with indices taken modulo 6 (so $T_6 = \{6, 1\}$); the jobs form a cycle in which consecutive jobs share exactly one tool. Its job–tool incidence matrix (rows are tools 1–6, columns jobs 1–6; a blank denotes 0) is

	1	2	3	4	5	6
t_1	1					1
t_2	1	1				
t_3		1	1			
t_4			1	1		
t_5				1	1	
t_6					1	1

Process the jobs in the cyclic order $1, 2, 3, 4, 5, 6$ and load by the rule of Section 2.2. The magazine evolves as

$$\{1,2,3\} \;\rightarrow\; \{1,2,3\} \;\rightarrow\; \{1,3,4\} \;\rightarrow\; \{1,4,5\} \;\rightarrow\; \{1,5,6\} \;\rightarrow\; \{1,5,6\},$$

inserting 3 tools to build the first magazine and then one tool at each of positions $3, 4, 5$: a switch cost of 6, equivalently 3 once the initial fill is not charged (Section 2.6). Section 3 shows that 6 is optimal, yet the two-phase heuristic of Section 2.3 returns 7. The 6-ring is thus the smallest instance on which grouping-then-sequencing is provably sub-optimal, and it recurs below as the canonical witness of a positive gap.

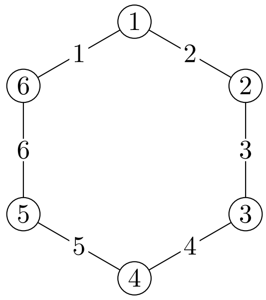

Figure 1: The 6-ring. Jobs 1–6 lie on a cycle; each edge is labelled by the single tool its two endpoints share ($T_j = \{j, j+1\}$). Adjacent jobs overlap in one tool; non-adjacent jobs are tool-disjoint.

2.2 The tool-loading oracle

The inner problem—optimal loading for a fixed order—is solved by a greedy rule whose optimality is classical.

Proposition 1 (32)**.** *For a fixed permutation π, an optimal magazine sequence is produced by the* Keep Tool Needed Soonest *(KTNS) policy: whenever a tool must be inserted into a full magazine, evict a tool whose next required use lies furthest in the future (evicting any tool never required again). KTNS runs in $O(n|T|)$ time and returns the minimum switch cost over all loading policies for π.*

A proof appears in Tang and Denardo [32]; see also Crama et al. [11]. KTNS reduces the SSP to a pure sequencing problem, $Z^* = \min_\pi \mathrm{KTNS}(\pi)$, and is the exact polynomial subproblem exploited by the Benders algorithm of Section 4.2.

Remark 1 (The loading subproblem is offline optimal caching)*.* For a fixed order the loading problem is exactly offline paging: the magazine is a cache of size b, position i requests the set $T_{\pi(i)}$, and an insertion is a cache miss. Proposition 1 is then the tool-switching form of Bélády's optimal replacement rule—evict the item whose next request is furthest in the future—which minimises misses on any fixed request sequence [3]. The contrast with the full problem is instructive: paging fixes the request order, whereas the SSP may *reorder* the jobs, and it is precisely this freedom that turns a polynomially solvable caching problem into an NP-hard one. Every method in this report is, at heart, a way to search over reorderings while delegating the loading to this exact oracle.

2.3 Grouping, sequencing, and the two-phase heuristic

A complementary viewpoint ignores order and asks only which jobs may *share* a magazine.

Definition 2 (Job Grouping Problem)**.** A *group* is a set of jobs $G \subseteq J$ whose combined tool requirement fits in the magazine, $|\bigcup_{j \in G} T_j| \leq b$. The Job Grouping Problem (JGP) partitions J into the minimum number K^* of groups; equivalently, it covers all jobs by the fewest tool configurations of size at most b.

The JGP is itself NP-hard and is solved exactly by the asymmetric-representatives formulation of Jans and Desrosiers [19] or by the set-covering column generation of Crama and Oerlemans [10]. It induces the classical two-phase heuristic for the SSP:

1. solve the JGP to obtain K^* groups and a configuration serving each;
2. solve the *Group Sequencing Problem*—order the configurations by a travelling-salesman-type problem whose transition cost is the number of tools that change, a small instance handled by standard solvers [1, 18]—to obtain a sequence of configurations;

3. concatenate the jobs group by group and evaluate the resulting order with KTNS.

We write H for the switch cost this heuristic returns. Since it produces a feasible schedule, $H \geq Z^*$ always; the objects of study are the *gap* $H - Z^*$ and the *ratio* H/Z^* (Section 3).

Example 2 (Grouping the 6-ring). On the running instance the JGP returns $K^* = 3$; one optimal grouping is $\{1,2\}, \{3,4\}, \{5,6\}$, served by the configurations $\{1,2,3\}$, $\{3,4,5\}$, $\{5,6,1\}$. Any two of these three configurations differ in two tools, so every ordering of the groups changes two tools at each of its two transitions, and the best schedule the heuristic produces costs $H = 7$ (free-initial 4). This is one more than the optimum of 6 (free-initial 3) attained in Example 1: a gap of exactly one. The discrepancy is not a failure of the sequencing phase—the groups are ordered optimally—but of the *grouping* phase, which insists on only $K^* = 3$ groups; Section 3 makes this precise.

2.4 The configuration–GTSP lens

Both the heuristic and the exact formulations live in the space of configurations.

Definition 3 (Configurations and clusters)**.** A *configuration* is a set $C \subseteq T$ with $|C| = b$. Let $\mathcal{C} = \{C \subseteq T : |C| = b\}$, and for each job j let $H_j = \{C \in \mathcal{C} : T_j \subseteq C\}$ be the *cluster* of configurations able to serve j. For configurations C, C' put $d(C, C') = |C' \setminus C|$, the number of tools inserted in passing from C to C'.

Restricting to full magazines ($|C| = b$) loses no optimality: retaining a tool that may be useful later never increases cost. Under this view a schedule is a walk $C_{(1)}, \dots, C_{(r)}$ in $\mathcal{C}$ that visits at least one configuration of every cluster H_j, with cost $\sum_l d(C_{(l)}, C_{(l+1)})$; the SSP is therefore a *generalised travelling-salesman problem with overlapping clusters* over $\mathcal{C}$ [8]. The clusters H_j overlap—a rich configuration serves many jobs—which is exactly what distinguishes the SSP from a textbook GTSP with disjoint clusters [14, 26, 28]. That a schedule and such a covering walk have equal cost is immediate in one direction—the distinct consecutive magazines of any schedule form a covering walk of the same cost—and the converse is established, in the stronger form ranging over *all* feasible groupings, as Theorem 1 of Section 3 (proof in Appendix A). This lens recurs throughout: it underlies the gap analysis of Section 3, the position-indexed formulations of Section 4.3, and the reading of the Benders master as a travelling-salesman skeleton (Section 4.2).

2.5 Two structural lower bounds

The following bounds are elementary but pervasive: they drive the gap theory of Section 3, calibrate the linear relaxations of Section 4, and serve as correctness checks for the computational study. Both are stated in the *free-initial* cost (Section 2.6), which does not charge the first magazine fill; the conversion to Definition 1 is the additive constant of Proposition 4.

Proposition 2 (Grouping bound)**.** $Z^* \geq K^* - 1$.

Proof. Fix any schedule and list its magazine states $M_1, \dots, M_n$; collapse each maximal run of equal states to a single representative, obtaining a sequence $C_{(1)}, \dots, C_{(r)}$ in which consecutive entries differ. Every job is served by some state, hence by some $C_{(l)}$, so $\{C_{(1)}, \dots, C_{(r)}\}$ is a family of sets of size at most b covering all jobs. Assigning each job to one configuration that contains its requirement partitions J into groups, one per *distinct* configuration, each group having combined requirement contained in that configuration and hence of size at most b; this is a feasible grouping, so the number of distinct configurations is at least K^*. Therefore $r \geq K^*$. Finally, each of the $r - 1$ consecutive transitions changes at least one tool, so the free-initial switch cost is at least $r - 1 \geq K^* - 1$. □

Proposition 3 (Coverage bound)**.** $Z^* \geq |U| - b$.

Proof. Each tool of U is required by some job and therefore occupies the magazine at some position, so it is inserted at least once unless it belongs to the initial fill, which contains at most b tools. Hence at least $|U| - b$ insertions occur in the free-initial count. □

Corollary 1. $Z^* \geq \max\{K^* - 1,\ |U| - b\}$.

The two bounds are individually tight on some families and simultaneously slack on others (Section 3). The coverage bound $|U| - b$ is exactly the value to which the multicommodity-flow relaxation of da Silva and Yanasse [12] collapses—a fact central to the assessment of Section 5.3. The two measure different things: on the 6-ring, $K^* - 1 = 2$ while $|U| - b = 6 - 3 = 3$, so the coverage bound is tight ($Z^* = 3$ free-initial, Example 1) and the grouping bound is slack, whereas on instances built from many nearly tool-disjoint groups the reverse holds. Neither dominates in general, and their maximum, though always valid, is not always attained: over 21,569 enumerated instances with jobs of sizes $\{2, 3\}$ at $b = 3$, 2,410 have Z^* strictly above both bounds, and the 8-job sliding-window ring at $b = 4$ ($T_j = \{j, j+1, j+2\}$ modulo 8, $|U| = 8$) has $Z^* = 5 > \max(3,\, 4)$.

2.6 Switch-cost convention

Formulations in the literature charge the initial magazine fill differently, and comparing their objective values naively produces spurious discrepancies; we therefore fix the convention.

Definition 4 (Empty-start and free-initial costs)**.** The *empty-start* cost charges every insertion from an initially empty magazine ($M_0 = \varnothing$, as in Definition 1). The *free-initial* cost does not charge the first fill; equivalently, it subtracts the $\min(b, |U|)$ tools loaded at the first position.

Proposition 4 (Convention identity)**.** *For every schedule,* $cost_{\text{empty}} = cost_{\text{free}} + \min(b, |U|)$.

Proof. An optimal loading (Proposition 1) keeps the magazine full, so the first position loads $\min(b, |U|)$ tools; the empty-start count charges these initial insertions while the free-initial count, by definition, omits them, and every later insertion is counted identically by both. □

The shift is a per-instance constant, so the gap $H - Z^*$ is convention-invariant whereas the ratio H/Z^* is not; ratio statements below are always taken in a single fixed convention. When objective values from different formulations are compared, they are first reduced to the empty-start convention via Proposition 4.

2.7 Related work

Exact formulations. Catanzaro et al. [6] give a family of compact integer formulations with a computational comparison, their strongest models adding overlap and linear-ordering inequalities. Laporte et al. [20] propose an integer program with a branch-and-cut. da Silva and Yanasse [12] give a multicommodity-flow formulation whose linear relaxation equals the coverage bound of Proposition 3. The configuration/GTSP formulation underlying our position-indexed models is developed by Colares and Wagler [8], Colares et al. [9].

Grouping and column generation. Crama and Oerlemans [10] introduce a set-covering formulation of the JGP solved by column generation with a strong relaxation; Jans and Desrosiers [19] give the asymmetric-representatives formulation. The pricing subproblem of the set-covering model is a Set-Union Knapsack Problem [17], NP-hard even under restrictive conditions. For branch-and-price we follow Barnhart et al. [2], Lübbecke and Desrosiers [21], Vanderbeck and Wolsey [33] and branch by the Ryan and Foster [31] same/different rule; subtour elimination uses the constraints of Miller et al. [23] with the lifting of Desrochers and Laporte [13], or lazily separated generalised cuts.

Adjacent exact studies. Two recent theses are close in method. Otiai [27] develops an exact branch-and-price for the Job Grouping Problem with Ryan–Foster branching and finds the set-covering relaxation strong enough to solve benchmark instances in seconds where the compact asymmetric-representatives model times out. Moreira [25] studies the SSP through a magazine-configuration formulation in the sense of Colares and Wagler [8] together with a configuration-generation heuristic, and measures the strength of that formulation's linear relaxation against the compact models; we revisit those measurements in Section 5.3.

Heuristics. Constructive and metaheuristic upper bounds include the branch-and-bound and bounding ideas of Ghiani et al. [16] and the hybrid genetic search of Mecler et al. [22], which seeds its population from JGP solutions.

The decomposition in practice. The group-then-sequence decomposition is not merely a theoretical device. In colour printing the SSP appears verbatim—ink cartridges are the tools, the cartridge rack the magazine—and Burger et al. [5] solve the industrial problem by exactly this decomposition, modelling the grouping as unicost set covering and the sequencing as a travelling-salesman problem, enumerating optimal groupings by Stirling-number arguments, and measuring the decomposition's suboptimality empirically: it is small on their instances, consistent with earlier reports they survey. What the literature does not contain is a *worst-case* analysis of the decomposition—bounds on its gap or ratio, extremal instances, or conditions for exactness; the approximation-flavoured results that exist for the SSP concern other heuristics. Section 3 supplies that analysis.

What remains open, and the route taken here. Three observations from the preceding paragraphs set the agenda. First, the decomposition heuristic has a strong empirical record but no worst-case theory; Section 3 builds one. Second, on the exact side the strongest compact relaxation equals the elementary coverage bound $|U| - b$ [12], and the configuration relaxation collapses to the same value on most instances [25]; progress therefore requires either a method that does not lean on the linear relaxation—the Benders route of Section 4.2, whose strength is an exact polynomial subproblem—or a formulation whose relaxation can exceed the bound, the position–transition route of Section 4.3. Third, the configuration view that is so effective for the JGP [27] lacks a tractable exact SSP counterpart, the natural model carrying exponentially many corrected subtour constraints (Section 4.3); that is the opening the position-indexed formulations address.

3 Structural analysis of the heuristic gap

The group-then-sequence heuristic is the standard practical method for the SSP, and where it has been tested against exact optima its measured suboptimality is small [5]; a worst-case analysis, however, does not exist. The source of the gap is that the heuristic commits to a *minimum-cardinality* grouping before it sequences, whereas the true optimum may be served by a larger grouping with cheaper transitions; the gap is precisely the difference between the two. This is what the running instance shows (Example 2), and the coverage bound $|U| - b$ that will limit the exact methods of Section 4 reappears here as the value that makes the gap vanish. Making the comparison exact lets us bound the gap tightly—it is the price of minimality, unbounded in absolute terms but within a factor b—record two sharper conjectures, and identify the algorithmic question, which grouping to sequence, that a better heuristic would have to answer.

3.1 The gap is the price of minimum-cardinality grouping

We attach a cost to a grouping by ordering its configurations optimally.

Definition 5 (Feasible grouping and its cost)**.** A *feasible grouping* is a partition $\mathcal{P} = \{G_1, \ldots, G_p\}$ of J together with, for each group, a configuration $C_l \supseteq \bigcup_{j \in G_l} T_j$ (which exists iff $|\bigcup_{j \in G_l} T_j| \le b$). Its *cost* is the cheapest free-initial walk through its configurations,

$$\gamma(\mathcal{P}) = \min_{\sigma \in S_p} \sum_{l=1}^{p-1} d\big(C_{\sigma(l)}, C_{\sigma(l+1)}\big).$$

The Job Grouping Problem seeks a feasible grouping of minimum cardinality K^*, and the two-phase heuristic evaluates γ on such a grouping. Removing the cardinality restriction makes the framework exact.

Theorem 1 (Grouping exactness)**.** $Z^* = \min_{\mathcal{P}} \gamma(\mathcal{P})$*, the minimum taken over* all *feasible groupings, of every cardinality.*

The proof (Appendix A) is a pair of constructions: any feasible grouping, processed group by group, is a schedule of cost $\gamma(\mathcal{P})$, giving "$\le$"; and any optimal schedule, its jobs grouped by the magazine under which they run, is a feasible grouping of no greater cost, giving "$\ge$".

Corollary 2 (The heuristic gap)**.** *Writing* $H = \min\{\gamma(\mathcal{P}) : |\mathcal{P}| = K^*\}$ *for the best two-phase cost,*

$$H - Z^* = \min_{|\mathcal{P}|=K^*} \gamma(\mathcal{P}) \;-\; \min_{\mathcal{P}} \gamma(\mathcal{P}) \;\ge\; 0,$$

and the gap is strictly positive exactly when no minimum-cardinality grouping attains the overall minimum of γ*.*

Remark 2 (A zero-gap certificate). Since $H \ge Z^* \ge \max(K^* - 1, |U| - b)$ (Corollary 1), any instance on which the heuristic attains a lower bound, $H = \max(K^* - 1, |U| - b)$, has zero gap. This already disposes of the k-rings with $k \in \{3, 4, 5\}$ (Table 1), where $H = |U| - b$.

Example 3 (Where the ring loses). The optimum of the 6-ring is realised by the *four*-configuration grouping $\{1,2,3\} \to \{1,3,4\} \to \{1,4,5\} \to \{1,5,6\}$, consecutive configurations differing in a single tool, for $\gamma = 3$ (cf. Example 1). Every minimum-cardinality grouping uses only $K^* = 3$ configurations and costs 4 (Example 2). The gap of one is precisely the difference in Corollary 2: the cheaper grouping is not of minimum cardinality, so the grouping phase never offers it to the sequencer.

3.2 The gap is real and unbounded

Corollary 2 locates the gap; the next question is whether it actually occurs, and how large it can grow. Table 1 collects the k-ring ($T_j = \{j, j+1\}$ modulo k, $b = 3$) for $k = 3, \ldots, 8$: the gap first appears at $k = 6$, where the ratio is largest. Replicating the worst ring makes the gap arbitrarily large.

Proposition 5 (Unbounded gap)**.** *Let* R_g *be* g *tool-disjoint copies of the* 6*-ring (*$b = 3$*,* $|U| = 6g$*). Then* $Z^* = 6g - 3$ *and* $H = 7g - 3$*, so the gap* $H - Z^* = g$ *is unbounded while the ratio* $(7g-3)/(6g-3)$ *decreases from* $4/3$ *at* $g = 1$ *towards* $7/6$*.*

The proof (Appendix A) pairs the coverage bound $Z^* \ge |U| - b = 6g - 3$ with a sequential schedule attaining it, and counts the heuristic's cost as g within-copy contributions of 4 and $g - 1$ inter-copy transitions of 3.

k	K^*	Z^*	H	gap	H/Z^*
3	1	0	0	0	—
4	2	1	1	0	1
5	3	2	2	0	1
6	3	3	4	1	4/3
7	4	4	5	1	5/4
8	4	5	6	1	6/5

Table 1: The k-ring ($b = 3$, free-initial costs).

Remark 3 (Perfectness does not predict the gap). One might hope that polyhedral regularity of an instance governs its gap; it does not. Form the *conflict graph* on the jobs, joining i and j when they cannot share a configuration ($|T_i \cup T_j| > b$). For the 6-ring this graph is the triangular prism, the complement of C_6, which is perfect, yet the gap is one; for the 5-ring it is the odd cycle C_5, which is imperfect [7], yet the gap is zero. Perfectness of the conflict graph and the size of the gap are therefore independent.

3.3 Worst-case analysis on general instances

Although unbounded in absolute terms, the gap is bounded in ratio—and everything in this subsection holds for *every* instance and every capacity, not only for the ring families that witness the extremes. Throughout, $q = |U| - b$, $K^* \geq 2$ (at $K^* = 1$ the heuristic is trivially exact, and then $q \leq 0$), and costs are free-initial.

Lemma 1 (Cost identity). *Restrict every magazine state to a subset of U (always possible when $K^* \geq 2$, and never worse). The free-initial cost of any full-magazine schedule then equals $q + R$, where $R \geq 0$ counts* re-insertions*: insertions of tools that were present earlier and evicted.*

Proof. Each of the q tools of U outside the initial load is required by some job, so it is inserted at least once; splitting all insertions into first-time ones (exactly q) and the rest (by definition R) gives the identity. □

Hence $Z^* = q + R_{\text{opt}}$ and $H = q + R_H$, so the gap $H - Z^* = R_H - R_{\text{opt}}$ is *exactly the excess re-insertions* that the minimum-cardinality grouping forces. The 6-ring's single extra switch is one such re-insertion: tool t_1, needed in the first and the last group but not in the middle one, is evicted at a group boundary and reloaded, where the optimal schedule keeps it resident throughout (Example 3).

Lemma 2 (Transition cap). *In any ordering of configurations contained in U, every transition inserts at most* $\min(b, q)$ *tools; hence* $H \leq (K^* - 1) \min(b, q)$.

Proof. A transition into C_{k+1} inserts $|C_{k+1} \setminus C_k| \leq |C_{k+1}| = b$ tools, and also at most $|U \setminus C_k| = q$ tools, since $C_{k+1} \subseteq U$ and $|C_k| = b$. A covering configuration inside U exists for every group, so some minimum-cardinality grouping uses only such configurations, and H is at most its cost. □

Theorem 2 (Worst-case guarantee). *For every instance with $K^* \geq 2$,*

$$\frac{H}{Z^*} \;\leq\; \frac{(K^* - 1) \min(b,\, q)}{\max(K^* - 1,\; q)} \;\leq\; \min(b,\; K^* - 1,\; q).$$

Proof. Divide Lemma 2 by $Z^* \geq \max(K^* - 1, q) \geq 1$ (Corollary 1). For the second inequality, $(K^* - 1)\, q / \max(K^* - 1, q) = \min(K^* - 1, q)$ and $\min(b, q) \leq q$ give the $\min(K^* - 1, q)$ part, while $(K^* - 1)/\max(K^* - 1, q) \leq 1$ and $\min(b, q) \leq b$ give the b part. □

This subsumes the immediate b-approximation ($H \le b(K^* - 1)$ divided by $Z^* \ge K^* - 1$): the guarantee tightens automatically on instances with few groups or with few tools in excess of the capacity.

Corollary 3 (Zero-gap classes, every b)**.** *(i) If $K^* = 2$ the gap is zero; indeed $H = Z^* = q$. (ii) If $|U| \le b + 1$ the gap is zero. (iii) If $K^* = 3$ then $H/Z^* \le 2$.*

Proof. All three are instances of Theorem 2, whose ratio bound is $K^* - 1 = 1$, $q \le 1$, and $K^* - 1 = 2$ respectively. For the equality in (i), pad the first group's configuration to size b with tools of the second group's requirement and pad the second configuration from the first; the single transition then inserts exactly the q tools of U missing from the first configuration, so $H \le q \le Z^* \le H$ by the coverage bound. □

Worst cases therefore require $K^* \ge 3$—consistent with every positive-gap instance found in enumeration having $K^* \in \{3, 4\}$.

The theorem's cap is far above the worst case actually observed. Exhaustive enumeration supports two sharper statements.

Conjecture 1 ($b = 3$ ratio)**.** For every instance with $b = 3$, $H/Z^* \le 4/3$.

Checked by exhaustive enumeration of the $b = 3$ two-tools-per-job family with $m \le 6$ (10,691 instances) and of mixed size-$\{2, 3\}$ jobs with $m \le 5$ (a further 6,065): the maximum ratio is exactly 4/3, attained at the 6-ring, with no counterexample. A second natural statement, suggested by every extremal instance known before this work and by enumeration over $b \in \{2, 3, 4\}$, would bound the gap by $K^* - 2$. It is true on wide regimes—for every $b \le 4$ at $K^* = 3$ in particular—but *false in general.* The results below give the exact heuristic cost at $K^* = 3$, quantitative bounds valid everywhere, the regimes where gap $\le K^* - 2$ does hold, and an explicit refutation just outside them.

Proposition 6 (Exact heuristic cost at $K^* = 3$)**.** *If $K^* = 3$ then, with configurations restricted to subsets of U,*

$$H \;=\; q \;+\; \min\; \min(x_{ab}, x_{bc}, x_{ca}), \qquad x_{uv} = |(C_u \cap C_v) \setminus C_w|,$$

the outer minimum over feasible 3-groupings and configuration choices. Hence the gap equals $\min\{\cdot\} - R_{\mathrm{opt}}$, and it vanishes exactly when some minimum-cardinality grouping admits configurations whose smallest pairwise overlap outside the third is at most R_{opt}.

Proof. The three configurations of a feasible 3-grouping are distinct (equal ones would merge two groups, contradicting $K^* = 3$) and cover U. For a path (C_a, C_b, C_c), splitting the cost into first-time insertions and re-insertions (Lemma 1) gives cost $q + |(C_a \cap C_c) \setminus C_b|$: the re-inserted tools are exactly those of C_c present in C_a but absent from C_b. Minimising over the three orderings selects the cheapest middle configuration, and H minimises over groupings and configuration choices. □

This turns grouping selection at $K^* = 3$ (Section 3.6) into an explicit criterion: seek the grouping minimising the smallest pairwise configuration overlap outside the third.

Lemma 3 (Overlap counting)**.** *Any three configurations $C_a, C_b, C_c \subseteq U$ of size b with $C_a \cup C_b \cup C_c = U$ satisfy $\min(x_{ab}, x_{bc}, x_{ca}) \le \lfloor (2b - q)/3 \rfloor$.*

Proof. Count tools by the number of configurations containing them: with a tools in exactly one configuration, $\sum x := x_{ab} + x_{bc} + x_{ca}$ in exactly two, and c_3 in all three, the tool count gives $a + \sum x + c_3 = b + q$ and the degree sum gives $a + 2\sum x + 3c_3 = 3b$; subtracting, $\sum x = 2b - q - 2c_3 \le 2b - q$. The minimum is at most the mean $\sum x/3$, and it is an integer. □

Proposition 7 (The gap at $K^* = 3$: quantitative bound and gap ≤ 1 regimes)**.** *Let $K^* = 3$ and $R_{\text{opt}} = Z^* - q$. Then*

$$\text{gap} \;\leq\; \max\Big(0,\ \min(q,\ \lfloor(2b-q)/3\rfloor) - R_{\text{opt}}\Big),$$

and the bound is attained by the 6-ring and by the $b = 4$ extremal witness. Consequently gap ≤ 1 *holds (i) for* every *instance with $b \leq 4$; (ii) whenever $2b \leq q + 3R_{\text{opt}} + 5$; (iii) whenever $R_{\text{opt}} \geq q - 1$; and (iv) at $Z^* = q = 3$ also when two of the four walk groups have combined requirement fitting one magazine. At $K^* = 3$ the bound* gap ≤ 1 *can therefore fail only in the corner $2b \geq q + 3R_{\text{opt}} + 6$ with $R_{\text{opt}} \leq q - 2$—necessarily $b \geq 5$—and Proposition 9 shows that there it does fail.*

Proof. The configurations of any feasible 3-grouping (one exists) cover U, so Proposition 6 with Lemma 3 gives $H \leq q + \lfloor(2b-q)/3\rfloor$, while Lemma 2 gives $H \leq 2\min(b, q) \leq q + q$; subtracting $Z^* = q + R_{\text{opt}}$ yields the display. For (i) at $b = 3$: $\lfloor(6-q)/3\rfloor \leq 1$ for all $q \geq 1$. At $b = 4$: the display is ≤ 1 for $q \geq 3$; for $q = 2$, either $R_{\text{opt}} \geq 1$ (display ≤ 1) or $Z^* = q \leq 2$, in which case the gap vanishes (Appendix A); $q \leq 1$ is Corollary 3(ii). Parts (ii) and (iii) read off the display; part (iv) is proved in Appendix A. ☐

Proposition 8 (A quantitative bound for every K^*)**.** *For every instance with $K^* \geq 2$,*

$$\text{gap} \;\leq\; \max\Big(0,\ \Big\lfloor \frac{q\,(b(K^*-1) - q)}{b+q} \Big\rfloor - R_{\text{opt}}\Big).$$

Proof. Fix a minimum-cardinality grouping; its configurations $C_1, \ldots, C_{K^*} \subseteq U$ are pairwise distinct (else two groups would merge) and cover U. In an ordering σ the path cost is $q + \sum_t(\text{runs}_\sigma(t) - 1)$, where $\text{runs}_\sigma(t)$ counts the maximal blocks of consecutive configurations containing t: every block start after the first is a re-insertion (Lemma 1). Let d_t be the number of configurations containing t, so $\sum_t d_t = K^*b$ over $b + q$ tools. Under a uniformly random ordering $\mathbb{E}[\text{runs}(t) - 1] = (d_t - 1)(K^* - d_t)/K^*$; this is concave in d_t, so by Jensen the expected total excess is at most $q\,(b(K^*-1) - q)/(b+q)$. Some ordering achieves at most the mean, the total is an integer, and subtracting $Z^* = q + R_{\text{opt}}$ gives the claim. ☐

At $K^* = 3$ this and Proposition 7 are incomparable, and both are tight at the 6-ring. The bound grows linearly in K^*; it is unconditional and, to our knowledge, the first quantitative worst-case bound on the decomposition valid for every K^*—on the $b = 5$, $K^* = 4$ extremal witness it gives 4 where the transition cap gives 8.

Corollary 4 (Small optima are gap-free)**.** *If $Z^* \leq 2$ then the gap is zero, for every b and every K^*.*

Proof. $Z^* \geq q$, so $q \leq 2$. If $q \leq 1$ the gap vanishes by Corollary 3(ii). If $q = 2$ then $Z^* = q$, and the walk argument of Appendix A yields at most $q + 1 = 3$ configurations, each serving a job, with $K^* \leq p$; for $K^* \leq 2$ the gap vanishes by Corollary 3(i), and for $K^* = 3$ necessarily $p = K^*$, which forces $H = Z^*$. ☐

Proposition 9 (The $K^* - 2$ bound fails in general)**.** *There are instances with $K^* = 3$ and gap $= 2$. One, with $b = 5$ and $U = \{0, \ldots, 8\}$:*

$$T = (\{0,1,2,4,7\},\ \{0,4,5\},\ \{1,5\},\ \{2,6,8\},\ \{3,4,5,8\}),$$

has $q = 4$, $Z^* = 4$ and $H = 6$ (both values exhaustive, reproduced by two independent implementations), hence gap $= 2 > K^* - 2$ *and $H/Z^* = 3/2$. The instance sits exactly on the boundary of the regimes of Proposition 7, $2b = q + 3R_{\text{opt}} + 6$, and attains that proposition's bound with equality:* $\min(q, \lfloor(2b-q)/3\rfloor) - R_{\text{opt}} = 2$.

Two consequences. First, the quantitative bounds of Propositions 7 and 8 are not way-stations towards $K^* - 2$: every known extremal instance—the 6-ring at $b = 3$, the $b = 4$ witness, and the $b = 5$ witnesses above—attains $\min(q, \lfloor(2b - q)/3\rfloor) - R_{\text{opt}}$ with equality, so whether this expression is the exact worst case at $K^* = 3$ is now the natural open question. Second, the ratio picture recalibrates: $H/Z^* = 3/2$ occurs already at $K^* = 3$, so worst-case ratios are not governed by K^* alone, and no uniform upper bound on the ratio (in b) is currently known.

Proposition 10 (The 4/3 bound holds whenever $K^* \leq 3$)**.** *Every $b = 3$ instance with $K^* \leq 3$ satisfies $H/Z^* \leq 4/3$. Conjecture 1 is therefore open only for $K^* \geq 4$.*

Proof. For $K^* \leq 2$ the gap is zero (Corollary 3). For $K^* = 3$, Proposition 7(i) gives gap ≤ 1, and a positive gap forces $Z^* \geq 3$ (Corollary 4), so $H/Z^* = 1 + \text{gap}/Z^* \leq 1 + 1/3 = 4/3$. ☐

The reduction to $K^* \geq 4$ is not vacuous: the edge-family observation that a positive gap forces $K^* = 3$ does *not* extend to mixed job sizes. Positive-gap $b = 3$ instances with $K^* = 4$ exist—for example $T = (\{0, 2, 5\}, \{0, 3, 6\}, \{1, 2\}, \{1, 4\}, \{1, 6\})$ with $Z^* = 4$, $H = 5$ and ratio 5/4—so an earlier draft's assumption to the contrary is false, and the $K^* \geq 4$ case is a genuine, populated regime. No ratio above 4/3 has appeared in it. Conjecture 1 carries no load elsewhere in this report: every other statement is proved unconditionally.

Remark 4 (The extremal curve in b). The constant 4/3 is specific to $b = 3$. The extremal ratio is non-decreasing in b: it reaches 4/3 already at $b = 4$ and 3/2 at $b = 5$, the latter now witnessed at $K^* = 3$ (Proposition 9) as well as at $K^* = 4$. Whether the supremum over all b stays below 2, or is bounded at all, is open; growth would require a shared bridge structure across unboundedly many groups, which the disjoint-copy family of Proposition 5 cannot provide—there the copies decouple and the ratio *decreases* to 7/6.

3.4 What determines the gap: grouping as an independence system

The results above are quantitative; this subsection asks from which combinatorial structure of an instance the gap can, even in principle, be read. The natural object is the family of feasible groups.

Definition 6 (Group complex and circuit clutter)**.** Let $\mathcal{F} = \{S \subseteq J : |\bigcup_{j \in S} T_j| \leq b\}$. Since $|T_j| \leq b$ and membership is preserved under taking subsets, $\mathcal{F}$ is an independence system on J; its inclusion-minimal non-members, the *circuits*, form a clutter $\mathcal{C}$.

Proposition 11 (K^* is a hypergraph chromatic number)**.** *A job set is a feasible group iff it contains no circuit; hence K^* equals the chromatic number of the circuit hypergraph $(J, \mathcal{C})$. The conflict graph of Remark 3 is exactly the family of two-element circuits, so $\chi(G_{\text{conf}}) \leq K^*$, strictly in general: for $T_1 = \{0, 1\}$, $T_2 = \{0, 2\}$, $T_3 = \{0, 3\}$ with $b = 3$ there are no conflicting pairs ($\chi = 1$), yet the triple is a circuit and $K^* = 2$.*

Proof. An infeasible set contains a minimal infeasible subset, and feasibility is downward closed, giving the first claim; colour classes of a proper colouring of $(J, \mathcal{C})$ are then exactly feasible groups. Two-element circuits are precisely the conflicting pairs, and every proper colouring of the hypergraph properly colours the graph. The star instance computes as stated. ☐

Proposition 12 (No matroid structure)**.** *$\mathcal{F}$ is not a matroid in general: for $T_1 = \{1\}$, $T_2 = \{2\}$, $T_3 = \{3, 4\}$ with $b = 2$, both $\{3\}$ and $\{1, 2\}$ are independent but neither $\{3, 1\}$ nor $\{3, 2\}$ is—the exchange axiom fails, so greedy and matroid-union arguments are unavailable without further hypotheses.*

Proof. Immediate: the relevant unions have sizes 2, 2, 3, 3. ☐

Proposition 13 (The circuit clutter does not determine the gap)**.** *Two instances with $b = 4$, $|U| = 7$, four jobs, isomorphic circuit clutters, and equal $K^* = 3$ and $Z^* = 3$ can have different gaps:*

$$I_0 : T = (\{0, 2, 6\}, \{1, 2, 3\}, \{2, 3, 4, 5\}, \{3, 4, 5\}), \qquad I_1 : T = (\{0, 1, 3, 6\}, \{1, 4, 5, 6\}, \{2, 4\}, \{3, 5\}).$$

In both, every job pair conflicts except the last two (the pairwise unions exceed b exactly there, and no larger circuit is minimal), yet I_0 has gap 0 and I_1 has gap 1.

Proof. The pairwise-union sizes are checked directly. Exhaustively over sequences and minimum-cardinality groupings, $Z^* = 3$, $H = 3$ for I_0 and $Z^* = 3$, $H = 4$ for I_1. □

Consequently no criterion reading only the circuit clutter—a fortiori only the conflict graph, strengthening Remark 3—can decide when the gap vanishes: tool-level structure is essential. That is precisely the level at which Proposition 6 operates; its overlap sizes x_{uv} are the data the clutter forgets. The clutter view retains its force on the *covering* side, where K^* itself lives (Proposition 11); the polyhedral questions this raises are set out in the planned work (Section 7).

3.5 Collapse under a setup cost

In practice each changeover incurs a fixed setup time beyond the per-tool exchanges. Model this by charging, in addition to the tool switches, a cost $\rho \geq 0$ per *configuration change*; the objective becomes switches $+ \rho \cdot$ (#configuration changes), where ρ is the ratio of setup time to unit switch time. A grouping into p configurations makes $p-1$ changes, so the setup term alone is minimised by the Job Grouping Problem; as ρ grows it dominates and pulls the optimum onto minimum-cardinality groupings.

Theorem 3 (Setup-cost collapse)**.** *If $\rho > H - Z^*$, then every optimal solution of the setup-augmented objective uses a minimum-cardinality grouping, and the two-phase heuristic is exact. Hence the collapse threshold satisfies $\rho_c \leq H - Z^*$; for the ring family $\rho_c = 1$.*

Proof. Every feasible grouping $\mathcal{P}$ has $|\mathcal{P}| \geq K^*$ and, by Theorem 1, $\gamma(\mathcal{P}) \geq Z^*$, so its augmented cost is $\rho(|\mathcal{P}| - 1) + \gamma(\mathcal{P}) \geq \rho(|\mathcal{P}| - 1) + Z^*$. The best minimum-cardinality grouping has augmented cost $\rho(K^* - 1) + H$. If $|\mathcal{P}| > K^*$, the former exceeds the latter whenever $\rho(|\mathcal{P}| - K^*) > H - Z^*$, which holds for all such $\mathcal{P}$ as soon as $\rho > H - Z^*$ (since $|\mathcal{P}| - K^* \geq 1$). Thus no over-cardinality grouping is optimal, and the heuristic—which selects the cheapest ordering of a minimum-cardinality grouping—attains the optimum. The ring value $\rho_c = 1$ follows from $H - Z^* = 1$ and the exact ring costs. □

Corollary 5. *On ring-like instances the setup-to-switch ratios $\rho \approx 3$–60 reported for real tooling [29, 30] far exceed ρ_c, so the two-phase heuristic is exactly optimal for the setup-augmented objective there.*

3.6 Which grouping to choose

Theorem 1 converts the heuristic's weakness into a concrete algorithmic question. Since the optimum is attained by *some* feasible grouping, and minimum-cardinality ones may miss it (Example 3), one should ask on *which* grouping to run the sequencing phase. Admitting a single extra group already recovers the ring optimum; in general the trade-off is between more configuration changes and cheaper individual transitions. Characterising the groupings that attain $\min_{\mathcal{P}} \gamma(\mathcal{P})$, or giving an efficient rule that improves on the minimum-cardinality choice, is open; the planned work returns to it (Section 7).

4 Exact methods

The gap analysis showed that $|U| - b$ is often the exact optimum; the question here is whether exact methods can exploit that, and what happens where the bound is loose. Two routes present themselves. Because the loading subproblem is polynomial (Proposition 1), a Benders decomposition can fix the order in a master and settle the loading exactly in the subproblem, generating cuts (Section 4.2). Alternatively the problem can be reformulated over configurations and solved by column generation—but the natural configuration formulation has exponentially many subtour constraints, and making it tractable is the obstacle that the position-indexed formulations of Section 4.3 resolve. We first record the compact formulations that serve as baselines.

4.1 Compact formulations and the coverage relaxation

Three compact formulations from the literature serve as the baselines of the computational study and as the yardstick against which relaxation strength is measured. Catanzaro et al. [6] give a family F1–F5 in job–position and ordering variables, the tighter members adding linear-ordering and overlap inequalities; we use F4 as a baseline. Laporte et al. [20] formulate the problem with tool-state variables and a branch-and-cut. da Silva and Yanasse [12] give a multicommodity-flow model in which each tool routes one unit of flow through the position network. The last is singled out by its relaxation.

Proposition 14. *The linear relaxation of the multicommodity-flow formulation equals the coverage bound $|U| - b$ of Proposition 3.*

Established by da Silva and Yanasse [12], and matched exactly by the position formulation's relaxation with counting rows (Proposition 16), this matters twice: the compact relaxation is no stronger than the elementary counting bound; and since $|U| - b$ equals Z^* on many families (Section 3), the model is fast exactly where that bound is tight and weak where it is not—a dichotomy quantified in Section 5.3.

4.2 A branch-and-Benders-cut algorithm

Idea. A schedule is an order together with a loading. Once the order is fixed, the minimum loading cost is computed exactly and in polynomial time by KTNS; so we place the *order* in an integer master and represent the loading cost by a single continuous variable θ, refined by cuts generated from the exactly solved loading subproblem. In contrast to column generation, whose pricing subproblem for this problem is NP-hard (Section 2.7), the Benders subproblem here is polynomial, so every cut is exact.

Master. Introduce a depot 0 and arc variables $x_{ij} \in \{0, 1\}$ for $i, j \in \{0, 1, \ldots, n\}$, $i \neq j$, with $x_{ij} = 1$ when job j is processed immediately after i (the depot marking start and end). The assignment constraints

$$\sum_{i \neq j} x_{ij} = 1 \quad (\forall j), \qquad \sum_{j \neq i} x_{ij} = 1 \quad (\forall i),$$

together with subtour-elimination constraints $\sum_{i,j \in S} x_{ij} \leq |S| - 1$ for $S \subseteq \{1, \ldots, n\}$, make the x a Hamiltonian order of the jobs. The master is

$$\min \ \theta \qquad \text{s.t.} \qquad \theta \geq \textstyle\sum_{i,j} w_{ij} \, x_{ij}, \quad \text{assignment, subtour, and Benders cuts,}$$

in which θ lower-bounds the loading cost and the subtour and Benders constraints are separated lazily.

An initial valid inequality. At the boundary between consecutive jobs i and j at least $w_{ij} := \max(0, |T_i \cup T_j| - b)$ tools are switched: the magazine holds at most b tools yet must serve first T_i and then T_j, so the members of $T_i \cup T_j$ absent at position i—at least $|T_i \cup T_j| - b$ of them—are inserted by position j. Summing over the chosen arcs gives the valid inequality $\theta \geq \sum_{ij} w_{ij} x_{ij}$, used to initialise the master.

Subproblem. Fix the order and relabel the jobs $1, \ldots, n$ along it. With $g_{t,i} \in [0, 1]$ indicating that tool t occupies the magazine at position i and $s_{t,i} \geq 0$ its insertion there, the loading cost is the optimal value of

$$\begin{aligned} \min \ & \sum_{t,i} s_{t,i} \\ \text{s.t. } & g_{t,i} = 1 && (t \in T_i), \\ & \textstyle\sum_t g_{t,i} \leq b && (\forall i), \\ & s_{t,i} \geq g_{t,i} - g_{t,i-1}, \quad s_{t,i} \geq 0 && (\forall t, i), \end{aligned}$$

with $g_{t,0} = 0$. The constraint matrix has the consecutive-ones property and is totally unimodular, so the relaxation is integral and, by Proposition 1, its optimum equals KTNS of the order. Writing R_i for the tool set of the job at position i, an optimal dual prices each requirement "tool t present at position i" by a multiplier $\bar{\alpha}_{t,i}$ and each capacity bound by $\bar{\beta}_i \geq 0$; linear-programming duality then yields the Benders optimality cut

$$\theta \ \geq \ \sum_i \sum_{t \in R_i} \bar{\alpha}_{t,i} \ - \ b \sum_i \bar{\beta}_i$$

(the multipliers of the unit upper bounds on the $g_{t,i}$ contribute a further term, suppressed here). The requirement multipliers carry the dependence on the order—through which job, hence which R_i, occupies each position—so the right-hand side is linear in the master's assignment variables; and because the dual feasible region does not depend on the order, the cut is globally valid, the defining property of a Benders cut.

Cut families. Two kinds of cut are separated, and the study ablates them. *(i) Benders optimality cuts* from the dual above, separated at integer master solutions and, optionally, at fractional ones (*fractional cuts*) to tighten the bound before branching. *(ii) Combinatorial KTNS cuts*: at an integer order $\bar{\sigma}$ with $\mathrm{KTNS}(\bar{\sigma}) = \bar{z}$, the inequality

$$\theta \ \geq \ \bar{z}\Big(1 - \sum_{(i,j) \in A(\bar{\sigma})} (1 - x_{ij})\Big)$$

is valid—it forces $\theta \geq \bar{z}$ when the order is exactly $\bar{\sigma}$ (all arcs $A(\bar{\sigma})$ selected) and is vacuous otherwise—and uses only the polynomial oracle, with no dual computation. Finally, *triplet bounds* generalise w_{ij} to ordered triples of jobs, contributing $O(n^3)$ valid inequalities that strengthen the root relaxation.

4.3 Position-indexed formulations

From the configuration model to a polynomial row set. The configuration formulation of Section 2.4, formalised by Colares and Wagler [8], is the natural exact model over configurations. As first stated, however, its subtour elimination is invalid; the correction of Moreira [25], through prize-collecting travelling-salesman constraints, restores exactness at the price of *exponentially many* constraints, one per subset of configurations. A Miller–Tucker–Zemlin reformulation [13, 23] is the standard device for removing such constraints, but here it trades one obstacle for another.

Proposition 15. *A per-configuration MTZ potential is exact but leaves one potential per configuration, hence exponentially many; aggregating the potentials to one per job is polynomial but no longer exact—it can exclude the optimum.*

The aggregated form fails under job domination: if $T_i \subseteq T_j$, every configuration serving j also serves i, so a transition between two such configurations places $\{i, j\}$ at both ends and forces the potentials of i and j into a two-cycle no ordering can satisfy; a four-job instance with $b = 3$ has optimum 1 that the aggregated model values at 2. Neither MTZ variant yields a tractable exact model. The position-indexed formulations below take a different route: they fix, in advance, a row set polynomial in $(n, |T|)$, so column generation only ever adds columns to existing rows, and they carry no subtour constraints at all.

Position–configuration formulation (PCF). Let $y_C^k \in \{0, 1\}$ indicate that configuration C occupies position k ($k = 1, \dots, n$) and let $w_t^k \geq 0$ count the insertion of tool t at position k:

$$
\begin{aligned}
\min\ & \sum_{t,k} w_t^k \\
\text{(P)}\quad & \sum_{C} y_C^k = 1, \quad k = 1, \dots, n; \\
\text{(Cov)}\quad & \sum_{k} \sum_{C \supseteq T_j} y_C^k \geq 1, \quad j \in J; \\
\text{(W)}\quad & w_t^k \geq \sum_{C \ni t} (y_C^k - y_C^{k-1}), \quad t \in T,\ k \geq 2; \\
\text{(T)}\quad & \sum_{k \geq 2} w_t^k + \sum_{C \ni t} y_C^1 \geq 1, \quad t \in U.
\end{aligned}
$$

The configuration variables y are generated as columns; the rows (P), (Cov), (W), (T) number $O(n|T|)$ and never change. Constraint (P) places one configuration per position, (Cov) serves every job, and (W) charges an insertion of t whenever it enters the magazine.

Proposition 16. *PCF is exact: its integer feasible points are exactly the SSP schedules, of equal cost. Its linear relaxation* without *(T) equals* 0*;* with *the counting rows (T) it equals* $|U| - b$.

The vanishing of the plain relaxation is the fractional pathology of Tang and Denardo [32]: a single convex blend of configurations, repeated at every position, satisfies (P), (Cov) and (W) at zero cost. The counting rows (T)—each tool of U is inserted at least once unless present at position one—restore the coverage bound, matching the multicommodity model (Proposition 14) with one extra family of rows.

Position–transition formulation (PTF). To exceed the coverage bound we index *transitions* rather than positions. Augment the configurations with an absorbing tool-less node $\perp$, used to pad schedules that change configuration fewer than $n - 1$ times, and let $z_{C,C'}^k \geq 0$ be a column for the step $k \to k+1$ that leaves configuration C and enters C'. The formulation is

$$
\begin{aligned}
\min\ & \sum_{k=1}^{n-1} \sum_{(C,C')} d(C, C')\, z_{C,C'}^k \\
\text{(S)}\quad & \sum_{(C,C')} z_{C,C'}^k = 1, \qquad k = 1, \dots, n-1; \\
\text{(F)}\quad & \sum_{(C,C')\colon t \in C'} z_{C,C'}^k = \sum_{(D,D')\colon t \in D} z_{D,D'}^{k+1}, \qquad t \in T,\ k = 1, \dots, n-2; \\
\text{(Cov)}\quad & \sum_{k=1}^{n-1} \sum_{(C,C')\colon T_j \subseteq C} z_{C,C'}^k + \sum_{(C,C')\colon T_j \subseteq C'} z_{C,C'}^{n-1} \ \geq\ 1, \qquad j \in J.
\end{aligned}
$$

Constraint (S) selects one transition per step; the flow-consistency rows (F) require, tool by tool, that what enters position $k+1$ at the end of step k is what leaves it at the start of step $k+1$, so the columns describe a single coherent walk $C^1 \to \cdots \to C^n$; and (Cov) serves every job at the configuration of some position—the tail of a step, or the head of the last. The objective charges $d(C, C') = |C' \setminus C|$ per transition. As in PCF the rows number $O(n|T|)$ and are fixed, while the transition columns are generated.

Proposition 17. *PTF is exact, and its linear relaxation is at least $|U| - b$ and can be strictly greater.*

On an instance whose optimum exceeds both bounds of Corollary 1, the PTF relaxation returns 2.10 against a coverage bound of 2; on the 6-ring it is tight at $Z^* = 3$. PTF is thus the first relaxation in this family to improve on the elementary bound, and it does so with a fixed, polynomial row set.

Symmetry reduction: the formulation PCF′. PCF carries a large *padding* symmetry. A schedule using $m < n$ distinct configurations is encoded by repeating configurations to fill all n positions, and the repeats may sit anywhere, so one physical schedule has many equal-cost encodings—distinct nodes to a tree search. We remove this freedom by relaxing the assignment equation (P) to an inequality and forcing the empty positions to a suffix:

$$(\mathrm{P}') \quad \sum_C y_C^k \le 1 \;\; (\forall k); \qquad (\mathrm{G}) \quad \sum_C y_C^{k+1} \le \sum_C y_C^k \;\; (k = 1, \ldots, n-1);$$

with (Cov), (W), (T) unchanged. Call this formulation PCF′. At an integer point the counts $\sum_C y_C^k$ are non-increasing in k, so the *used* positions form a prefix $1, \ldots, p$ and the empties a suffix, with the cost-free first configuration at position 1. The contiguity rows (G) are not decorative: without them an interior empty position would make (W) read the re-entry as a full reload and (T) misattribute the free load, so (G) is what keeps the relaxed model faithful.

Proposition 18. *PCF′ is exact, and its linear relaxation equals that of PCF—$|U| - b$ with the counting rows (T), and 0 without. It removes symmetric integer encodings but does not strengthen the bound.*

The proof is in Appendix A. The separation of concerns is deliberate: PCF′ attacks symmetry, and only PTF attacks the bound.

Column generation and robust branching. Both PCF′ and PTF have $O(n|T|)$ rows but exponentially many columns ($\binom{|T|}{b}$ configurations, or arcs), so their relaxations are solved by *column generation*: a restricted master over a small column pool is reoptimised, and a pricing subproblem repeatedly supplies a column of most-negative reduced cost until none remains. Driving this at every node of a branch-and-bound tree gives *branch-and-price* [2, 21]. Branching on a single column y_C^k is ineffective under the position symmetry and forces the pricer to carry a growing list of forbidden columns. We branch instead on the *presence* variables $a_t^k = \sum_{C \ni t} y_C^k \in \{0, 1\}$ already appearing in (W): fixing a_t^k to 0 or 1 forbids or forces tool t at position k, which in the pricing subproblem merely shifts one dual weight and leaves the oracle's *form* unchanged. This *robust* branching keeps pricing and branching compatible at every node.

Pricing PCF′: a b-subset with coverage bonuses. Let $\alpha_k, \gamma_k \le 0$, $\pi_j, \mu_{t,k}, \lambda_t \ge 0$ be the duals of (P′), (G), (Cov), (W), (T). Direct dual accounting (collecting the coefficient of y_C^k, which enters (W) and (T) through $a_t^k = \sum_{C \ni t} y_C^k$) gives the reduced cost

$$\bar{c}(C, k) = \beta_k - \sum_{t \in C} \rho_t^k - \sum_{j:\, T_j \subseteq C} \pi_j, \qquad \rho_t^k = -\mu_{t,k}[k \ge 2] + \mu_{t,k+1}[k \le n-1] + \lambda_t\,[k = 1,\; t \in U],$$

with β_k depending only on the position. For each k the most-negative reduced cost is obtained by maximising the configuration-dependent part,

$$\max_{|C|=b} \sum_{t\in C} \rho_t^k + \sum_{j:\, T_j\subseteq C} \pi_j,$$

i.e. choosing b tools to collect per-tool rewards plus a bonus for each job whose *entire* requirement lands inside C. This is a Set-Union Knapsack Problem [17]—the grouping-pricing class of Crama and Oerlemans [10]—solved by enumerating the $\binom{|T|}{b}$ subsets at magazine sizes, or by a compact mixed-integer program. The branching dual of a_t^k enters additively in ρ_t^k, so branching does not change the oracle.

Pricing PTF: a coupled pair of subsets. For PTF a column is an arc (C, C'), and its reduced cost contains the bilinear term $\sum_t [t \in C' \setminus C]\,(1 - \lambda_t)$ together with chaining-dual terms on the tools of C and C'. The pricer cannot therefore separate into a single set: it must choose *two* b-subsets $C \neq C'$ at once, a bilinear selection linearised by McCormick on the products $x_t'(1 - x_t)$ and solved as a mixed-integer program with $2|T|$ binaries. This heavier pricing is the algorithmic price of PTF's stronger bound. The contrast—PCF$'$ pairing a weak bound with a cheap single-set pricer, PTF a stronger bound with a costly coupled-pair pricer—is the central trade-off assessed in Section 5.3.

5 Computational study

The methods of Section 4 have been implemented and are being benchmarked. We report the experimental design; the large-scale comparison is in progress, and its tables are the report's only deferred content.

5.1 Experimental design

The benchmark draws on the standard SSP instance families of Catanzaro et al. [6] ($n = 8$–15 jobs, $m = 5$–15 tools, $b = 3$–5), Crama et al. [11] ($n = 10$, $m = 10$, $b = 4$) and Laporte et al. [20] ($n = 8$–15, $m = 10$–15, $b = 4$–5), spanning a range of tool densities and capacities. Three things are measured. *(i)* The branch-and-Benders-cut solver is run as a full-factorial ablation of its cut families—combinatorial against dual optimality cuts, fractional separation on or off, triplet bounds on or off—against the three compact baselines. *(ii)* The position-indexed branch-and-price is run in its PCF$'$ and PTF variants. *(iii)* For every solver and instance the *root linear-relaxation value* is recorded alongside the optimum where the solver exposes it reliably, so that bound tightness can be studied directly. The common metric is the empty-start KTNS cost of the returned sequence, well defined for every method regardless of its native objective. Runs are pinned to a fixed solver thread count per task for reproducible timing, with a per-instance time limit, and are sharded across a compute cluster.

5.2 Results

The campaigns are executing at the time of writing. The completed study will report, across the instance families: solve rates and running times for each method; the marginal contribution of each cut family in the branch-and-Benders-cut ablation; the comparison of the position-indexed branch-and-price against the compact baselines on the shared metric; and the distribution of the root-relaxation gap, which the following subsection argues is the decisive quantity.

5.3 The competitiveness of configuration branch-and-price

A guiding question for the exact methods is whether a configuration-based branch-and-price can be made fast for the SSP, and the structural results already constrain the answer. By Propositions 14 and 16 the configuration relaxation collapses to the coverage bound $|U| - b$, and by the gap theory of Section 3 this bound equals Z^* on a large part of the instance space. Where the root bound already equals the optimum branching is unnecessary; where it does not, a relaxation no stronger than $|U| - b$ forces a search tree that faster pricing or branching cannot avoid. The contrast with the Job Grouping Problem is telling: there the set-covering relaxation is strong enough that an analogous branch-and-price closes benchmark instances in a handful of nodes [27], whereas for the SSP the corresponding configuration relaxation does not improve on the trivial bound on most instances [25]. A configuration branch-and-price for the SSP is therefore *bound-limited* rather than implementation-limited. The position–transition formulation, the first in the family to exceed $|U| - b$ (Proposition 17), is the structural lever this diagnosis points to; but it pays off only where the gap opens, and broadening that improvement is part of the planned work (Section 7).

6 Conclusions

6.1 Summary

A single device—the equivalence between the SSP and a generalised travelling-salesman problem over tool configurations—has organised both a structural theory of the standard heuristic and a family of exact methods. On the structural side we proved that admitting arbitrary groupings makes the configuration framework exact (Theorem 1), so that the two-phase heuristic's gap is exactly the price of a minimum-cardinality grouping (Corollary 2); we bounded that gap from below and above (Corollary 1, Theorem 2), isolated zero-gap classes valid for every capacity (Corollary 3), showed the gap can grow without bound (Proposition 5), and exhibited a setup-cost regime in which it vanishes identically (Theorem 3). Of the two natural conjectures, the 4/3 ratio at $b = 3$ is now proved whenever $K^* \leq 3$ (Proposition 10) and open only at $K^* \geq 4$; the gap bound $K^* - 2$ is proved on wide regimes and *refuted* outside them by an explicit counterexample attaining our quantitative bounds with equality (Propositions 7 and 9), with the heuristic cost at $K^* = 3$ given in closed form (Proposition 6). The circuit clutter of the grouping system computes K^* yet provably does not determine the gap (Section 3.4). On the algorithmic side we gave a branch-and-Benders-cut solver whose subproblem is exact and polynomial, and position-indexed formulations with a fixed row set, among which the position–transition model is the first configuration relaxation to improve on the coverage bound. The methods are implemented; their large-scale comparison is under way.

6.2 Beyond the project horizon

The near-term programme is set out in Section 7; three further directions fall outside it and are recorded here for the sequel.

The position–transition device beyond the SSP. Nothing in the position–transition construction is specific to tool switching except the transition metric: position-transition columns with per-element consistency rows apply to any generalised travelling-salesman problem with overlapping clusters [28]. Whether the relaxation strength observed here persists in that generality is open, and an affirmative answer would carry the contribution beyond the SSP.

Approximation and online variants. The ratio theory of Section 3 should be reconciled with worst-case approximation guarantees reported elsewhere for the SSP, read critically as to the bound and convention against which they hold. Separately, the caching equivalence of Remark 1 suggests an online SSP, in which jobs arrive over time, analysed competitively against paging machinery—with the offline loading oracle as a natural baseline.

Learning-augmented column generation. Machine-learning acceleration of pricing and branching [4, 15, 24] is an active but crowded field. It is worth pursuing here only if the benchmark data expose a concrete learning signal—such as the pronounced imbalance between zero-gap and positive-gap instances that the structural results predict—rather than as an end in itself.

7 Planned work

The preceding sections report what has been done; this section sets out what remains, in the order in which it will be carried out. Firm dates are deliberately not attached—an extended mid-summer break falls within the remaining period—but the phasing is fixed by dependency: each phase consumes the output of the one before it. Phase 1 is expected to close before the break, and the aim is to complete the programme ahead of the project's formal end.

Phase 1 (under way): complete the computational campaigns. Two campaigns are running: the full-factorial cut ablation of the branch-and-Benders-cut solver, and the position-indexed branch-and-price in its PCF′ and PTF variants, both against the compact baselines and both recording root relaxations alongside optima (Section 5.1). The deliverable is the completed tables and their analysis—solve rates, running times, the marginal value of each cut family, and the distribution of root gaps. This phase comes first because what follows it is steered by its bound-tightness data.

Phase 2: the exact extremal gap law. With the K^*-2 bound refuted (Proposition 9), the standing question is the true extremal law. Every known extremal instance attains $\min(q, \lfloor(2b-q)/3\rfloor)-R_{\text{opt}}$ (Proposition 7); deciding whether that expression is the exact worst case at $K^*=3$, tightening Proposition 8 at $K^* \geq 4$ (derandomising the ordering, sharpening the degree profile), and settling the 4/3 conjecture in its one remaining case $K^* \geq 4$ (Proposition 10 covers $K^* \leq 3$) are the targets, with enumeration directed at the boundary regimes where the counterexamples live. In parallel, the exact expression of Proposition 6 turns grouping selection (Section 3.6) at $K^*=3$ into a concrete criterion, to be tested against the enumerated pool of optimal groupings.

Phase 3: strengthen the position–transition relaxation; the polyhedral view. Where Phase 1 finds the coverage bound loose, attention turns to PTF: characterising the instances on which its relaxation exceeds $|U|-b$ (Proposition 17) and deriving valid inequalities that widen the margin. The polyhedral thread pursued with Prof. Wagler now rests on the results of Section 3.4: the circuit clutter computes K^* but provably cannot decide the gap (Proposition 13), so the productive questions are on the covering side—whether idealness of the covering system characterises the instance classes on which the JGP relaxation is integral, and whether the blocker's minimal obstructions localise the bridge tools that generate re-insertions.

Phase 4: consolidation. The closing phase assembles the results into the final report and shapes the strongest thread into a manuscript for submission.

A Proofs

Proof of Theorem 1. Both inequalities are explicit constructions; by Proposition 1 we may assume the optimal schedule loads by KTNS, so every magazine is full, $|M_i| = b$.

($\leq$). Let $\mathcal{P} = \{G_1, \ldots, G_p\}$ be a feasible grouping with configurations $C_1, \ldots, C_p$, and let σ attain the minimum in Definition 5. Process the groups in the order $G_{\sigma(1)}, \ldots, G_{\sigma(p)}$, and within each group its jobs in any order, holding magazine $C_{\sigma(l)}$ throughout group $G_{\sigma(l)}$. This is feasible, since each $j \in G_{\sigma(l)}$ satisfies $T_j \subseteq \bigcup_{i \in G_{\sigma(l)}} T_i \subseteq C_{\sigma(l)}$ and $|C_{\sigma(l)}| = b$. The magazine is constant within a group and changes only at the $p-1$ group boundaries, so the free-initial switch cost is $\sum_{l=1}^{p-1} d(C_{\sigma(l)}, C_{\sigma(l+1)}) = \gamma(\mathcal{P})$. Hence $Z^* \leq \gamma(\mathcal{P})$, and minimising over $\mathcal{P}$ gives $Z^* \leq \min_{\mathcal{P}} \gamma(\mathcal{P})$.

($\geq$). Let $(\pi, \mathbf{M})$ be an optimal schedule with full magazines. Partition J into the maximal blocks of consecutive positions sharing one magazine; let these be $B_1, \ldots, B_r$ in order, with (size-b) magazines $D_1, \ldots, D_r$ and $D_l \neq D_{l+1}$. Each block is a group whose jobs run under D_l, so $\bigcup_{j \in B_l} T_j \subseteq D_l$ with $|D_l| = b$; the blocks thus form a feasible grouping $\mathcal{P}'$ with configurations $D_1, \ldots, D_r$. The identity ordering of these configurations is one admissible σ, so $\gamma(\mathcal{P}') \leq \sum_{l=1}^{r-1} d(D_l, D_{l+1})$, which is exactly the free-initial switch cost of $(\pi, \mathbf{M})$, namely Z^*. Therefore $\min_{\mathcal{P}} \gamma(\mathcal{P}) \leq \gamma(\mathcal{P}') \leq Z^*$. Combining the two inequalities gives equality. $\square$

Proof of Proposition 5. The copies use pairwise disjoint tool sets, so $|U| = 6g$ and Proposition 3 gives $Z^* \geq 6g - 3$. For a matching schedule, process the copies one after another, each in the cyclic order of Example 1. Within a copy that loading makes 3 insertions to build the first magazine and 3 more across the copy; since consecutive copies are tool-disjoint, entering a new copy rebuilds its first magazine afresh (3 insertions). The cost is 6 insertions for the first copy and 6 for each of the others, i.e. $6g$ in the empty-start count and $6g - 3$ free-initial; hence $Z^* = 6g - 3$.

For the heuristic, a minimum-cardinality grouping of R_g uses $K^* = 3g$ configurations, three per copy. In the ring, those three configurations are pairwise at distance 2, so any ordering of one copy's three contributes $2 + 2 = 4$; and any transition between configurations of different copies costs 3, the copies being tool-disjoint. Connecting g copies needs at least $g - 1$ inter-copy transitions, so every ordering costs at least $4g + 3(g-1) = 7g - 3$, attained by keeping each copy's configurations consecutive. Thus $H = 7g - 3$ and $H - Z^* = g$. $\square$

Proof of Proposition 7: the walk-based claims. Suppose $Z^* = q$ and take an optimal schedule; its walk of distinct configurations $D_1, \ldots, D_p$ may be assumed, after excising configurations serving no job (the transition cost d obeys the triangle inequality $d(A, C) \leq d(A, B) + d(B, C)$, and no covering walk costs less than q), to have every configuration serve a job. With zero re-insertions every insertion is first-time, so the $p - 1$ transitions insert q tools in total and $p \leq q + 1$; assigning each job to a serving configuration exhibits a feasible p-grouping, so $p \geq K^*$; and if $p = K^*$ that grouping is minimum-cardinality of cost Z^*, forcing $H = Z^*$. In particular, $Z^* = q \leq 2$ gives $p \leq q + 1 \leq 3 = K^*$, so the gap vanishes—the claim used in part (i).

For (iv), let $q = 3$; if $p = 3$ the gap vanishes as above, so let $p = 4$. Each transition inserts exactly one fresh tool: $D_{l+1} \setminus D_l = \{x_{l+1}\}$ and $D_l \setminus D_{l+1} = \{e_{l+1}\}$; and zero re-insertion forces *persistence*: a tool of $D_i \cap D_j$ ($i < j$) belongs to every intermediate D_l. Write U_l for the requirement of group G_l and suppose $|U_i \cup U_j| \leq b$ for some $i < j$. Merging G_i and G_j, keeping the other two groups with their walk configurations, yields a minimum-cardinality grouping; it remains to choose its configuration $C \supseteq U_i \cup U_j$ and an ordering whose wrap term $|(\text{first} \cap \text{last}) \setminus \text{middle}|$ (Proposition 6) is at most one, giving $H \leq q + 1 = Z^* + 1$. For $(1, 2)$: take $C \subseteq D_1 \cup D_2$ (possible, the union has size $b{+}1$) and order (C, D_3, D_4); the wrap lies in $(D_1 \cap D_4 \setminus D_3) \cup (D_2 \cap D_4 \setminus D_3) \subseteq \varnothing \cup (D_2 \setminus D_3) = \{e_3\}$, using persistence. For $(3, 4)$: order (D_1, D_2, C) with $C \subseteq D_3 \cup D_4$; the wrap lies in $D_1 \cap (D_3 \cup D_4) \setminus D_2 = \varnothing$ by persistence. For

$(1,3)$, $(2,4)$, $(1,4)$: the orderings (C, D_2, D_4), (D_1, D_3, C), (C, D_2, D_3) bound the wrap by $\{x_3\}$, $\{e_3\}$, $\{x_3\}$ respectively, by the same two facts. For $(2,3)$: the set $W = U_2 \cup U_3 \cup (D_1 \cap D_4)$ lies in $D_2 \cup \{x_3\}$ (persistence places $D_1 \cap D_4$ inside $D_2 \cap D_3$), so $|W| \leq b{+}1$ and a configuration $C \supseteq U_2 \cup U_3$ inside W omits at most one element of $D_1 \cap D_4$; the ordering (D_1, C, D_4) has wrap $|(D_1 \cap D_4) \setminus C| \leq 1$. □

Proof of Proposition 18. Exactness. Take an optimal schedule and collapse equal consecutive magazines to a trajectory $D_1, \ldots, D_m$ ($m \leq n$) as in the proof of Theorem 1. Set $y^l_{D_l} = 1$ for $l = 1, \ldots, m$, leave positions $m{+}1, \ldots, n$ empty, and $w^k_t = \max(0, a^k_t - a^{k-1}_t)$. Then (P′) holds ($= 1$ on the prefix, $= 0$ after) and (G) holds (the prefix is contiguous), while (Cov), (W), (T) hold exactly as for PCF; the full-to-empty step $m \to m{+}1$ and all later steps add no cost, so the objective is $\sum_{l<m} d(D_l, D_{l+1}) = Z^*$. Hence the optimum is at most Z^*. Conversely, any integer-feasible PCF′ point has, by (G), its used positions in a prefix $1, \ldots, p$, each holding one configuration D_k; by (W) the objective is at least $\sum_{k=2}^{p} d(D_{k-1}, D_k)$, and by (Cov) every job is served by some D_k, so assigning each job to a covering position yields a schedule of cost at most the objective. Hence the optimum is at least Z^*.

LP invariance. Every PCF-feasible point has $\sum_C y^k_C = 1$ for all k and so satisfies (P′) and (G); the PCF′ relaxation thus minimises over a superset and is no larger than PCF's. For the reverse, summing (T) over $t \in U$ and using $\sum_{t \in U} a^1_t \leq \sum_{t \in T} a^1_t = b \sum_C y^1_C \leq b$ (the last bound now from (P′) rather than (P)) gives objective $\geq |U| - b$ at every PCF′ point, so the two relaxations coincide wherever PCF's value is $|U| - b$. The plain relaxation, dropping (T), is certified 0 by the same blended configuration used for PCF. □

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